UNIVERSITY OF SURREY

EEEM073 — AI and Sustainability

Spatiotemporal Wildfire Severity Forecasting and Optimisation for Sustainable Forest Management

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SDG Alignment	Goal
SDG 13 — Climate Action	Predicting and mitigating fire-driven carbon emissions
SDG 15 — Life on Land	Protecting Central African savanna ecosystems from wildfire

1. Introduction

1.1 Problem Definition and Motivation

Since the 1990s, global warming rising approximately 35% have resulted in climate-related disasters with an increase in both frequency and intensity events like extreme floods, hurricanes, wildfires, and droughts[15]. Not only these occurrences have caused enormous human but also economic losses , the 10 biggest weather disasters of 2025 alone caused an estimated $120 billion in damages and disproportionately affect low-income nations [16]. [1, 2]

Among all disasters, wildfires stand out as one of the most damaging. The total area of land burned globally has grown by an estimated 25% over the past two decades. Central Africa is the focus of this study since it is responsible for 60–70% of all fire-related carbon emissions worldwide making it the single most important region when it comes to understanding how wildfires effect climate change.

Today, forest fire suppression in Central Africa is primarily reactive: firefighters respond after a fire has already ignited.Without an advance notice regarding whether and how crucial flames will burn, it is impossible to pre-position fire-fighting teams, tools, or execute controlled burns in time. In 2022, fires in this area released about 420 million tons of carbon dioxide, more than the complete annual emissions of many industrialized countries [1, 17]. This project builds an AI machine that can predict the total area that will potentially burn each month, giving groups the lead time they want to operate before the fireplace starts as opposed to after.

Two aims were taken from United Nations Sustainable Development Goals: **SDG 13 (Climate Action)** by providing tools to reduce fire-driven carbon emissions through early intervention and **SDG 15 (Life on Land)** by protecting terrestrial ecosystems and biodiversity from uncontrolled fire spread , which became my motivation.

1.2 Related Work

Researchers have tried to predict wildfires in many different ways. Early approaches relied on weather-based fire danger indices essentially risk scores calculated from temperature, wind, and humidity but these tools are too coarse to guide local decision-making. In recent years, machine learning methods have shown much greater promise. Studies inside the Mediterranean and Greece have established that AI models trained on satellite data can anticipate burned areas larger as it should be than traditional methods [7, 11]. Others have used image-recognition strategies to model fire spread across landscapes [12].

Although Central Africa is the immediate environment responsible for the most emissions from fireplaces worldwide, it has received much much less research interest than North America, Europe or Australia. Existing models also rarely incorporate physical rules for example, the constraint that a fire cannot burn more land than actually exists and few studies compare multiple AI approaches on the same dataset. This project fills both gaps by applying and comparing three AI models specifically to Central African fire data, while ensuring all predictions remain physically realistic.

1.3 Research Questions

This project addresses the following primary research questions:

- Can a standard decision tree AI model give a useful first estimate of how much land will burn each month, using only information about vegetation cover types?
- Does an AI model that “remembers” the past 12 months of fire history predict significantly better than one that treats each month in isolation?

- Can an AI model that reads the landscape as a spatial map rather than as a timeline predict fire severity just as well?
- Can the AI be designed to respect physical common sense for example, never predicting that more land burns than physically exists while not compromising accuracy?
- Can the best-performing AI model be made sustainable enough to run on low-cost hardware in the field, without sacrificing meaningful accuracy?

1.4 Objectives

This project has five practical goals:

- Build and compare multiple AI models for predicting monthly wildfire severity, using a large satellite dataset (ESA Fire_cci v5.1) covering Central Africa from 2017 to 2022.
- Use fire history data to capture the seasonal patterns of burning for example, the predictable peak of fires each July–October and how fire seasons vary from year to year.
- Use explainability techniques to show which vegetation types and landscape features drive fire severity providing decision-makers with transparent, evidence based guidance rather than a "black box" output.
- Reduce the dimensions and power intake of the high-quality modes in an effort to run on low-cost, low-power hardware that can be deployed in rural areas of Central Africa.
- Measure the carbon emissions created by running each AI model, and show that the most compressed version provides the optimal balance of accuracy versus environmental cost.

1.5 Dataset Overview

The data used in this project comes from the [1, 18] European Space Agency's Fire_cci v5.1 product, a satellite-based record of burned land area covering the entire globe from 2001 to 2022. The satellite (NASA's MODIS) measures how much land has burned each month in grid squares roughly 28 km across. A comprehensive and consistent view of firefighter behavior has been captured over a period of 20-plus years. For the purposes of this project, only data collected between 2017 and 2022 has been used, with emphasis on the most current data reflecting changing patterns. Data was further cropped to Central Africa, which contains the highest concentration of flame activity within the context of this project (Congo Basin, Miombo Woodlands, northern savanna belt) and the region where the need for improved forecasting tools is the greatest.

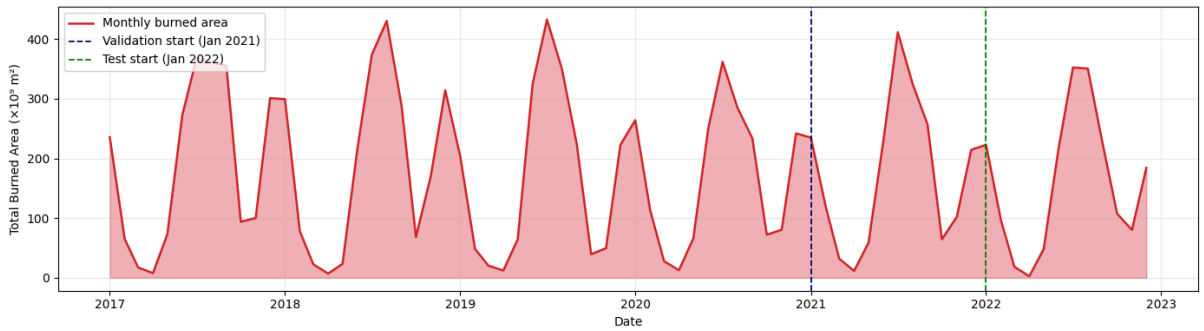

Figure 1: Monthly Total Burned Area in Central Africa (2017–2022)

The area in red highlights the training period for the analysis. The dataset has over 74 billion data points which spans globally with a total land area (720 x 1440 grid cells) and 72 monthly time steps. This creates a much more sophisticated data set than traditional tabular formatted data in CSV format necessitating specialized multi-dimensional tools (xarray, Dask) due to memory restrictions during processing.

Variable	Unit	Description
burned_area	m²	Total burned area per 0.25° cell per month. Primary regression target.
burned_area_in_vegetation_class	m²	Burned area disaggregated across 18 land-cover types. Primary feature set.
fraction_of_burnable_area	unitless [0–1]	Proportion of cell physically capable of burning. Used for masking.
fraction_of_observed_area	unitless [0–1]	Proportion of cell with valid MODIS observations (cloud-free).
standard_error	m²	Measurement uncertainty. Excluded from features (circular dependency).
vegetation_class_name	string	Human-readable label for each of the 18 land-cover classes.

Table 1: Data Dictionary — ESA Fire_cci v5.1 Key Variables

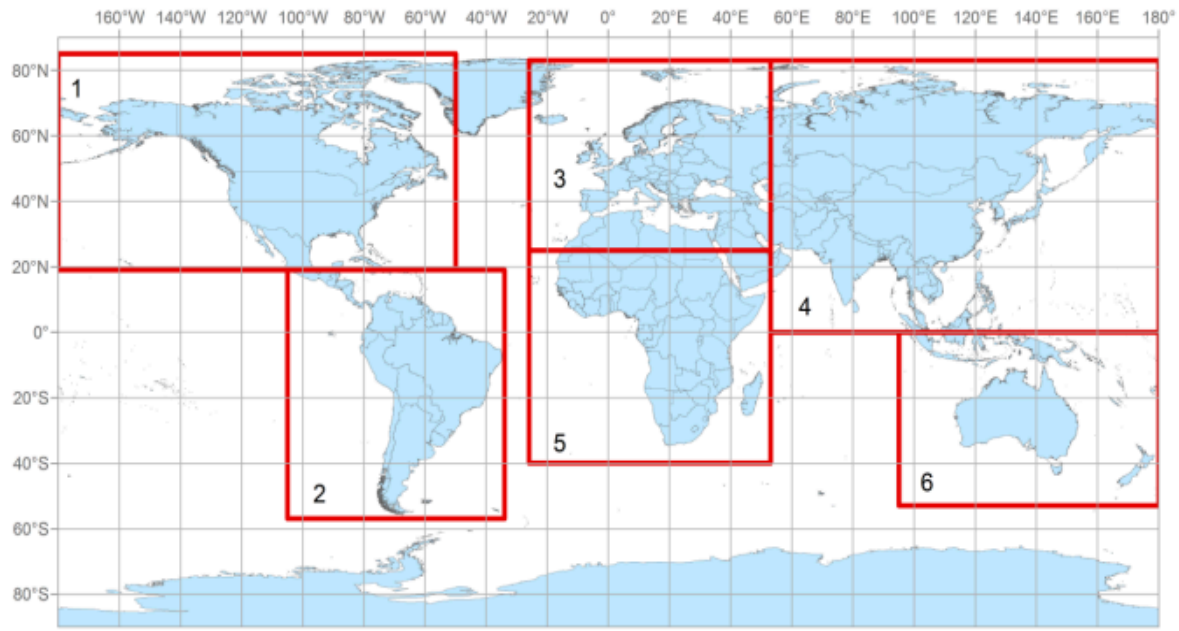

Figure 2: Geographical distribution of the dataset [18]

The geographic scope of subset 5 (Fig 2) was even more cropped to Central Africa (10°N to 20°S, 10°E to 45°E) to concentrate modelling resources on the highest fire-density region and reduce the global pixel sparsity from 98.8% zeros to approximately 14% zeros insulating the model from bias. This area is divided into three main fire biomes of sub-Saharan Africa: the northern savanna corridor, the Miombo woodland belt, and the Congo Basin.

2. Data Understanding, Pre-processing and Exploration

2.1 Data Quality Assessment

When we first examined the full global dataset, we found that 98.8% of all locations recorded no fire in any given month; in other words, almost everywhere on Earth does not burn most of the time. To make the AI focus only on areas where fire is physically possible, we removed locations that can never burn: open ocean, permanent ice sheets, and bare rock. We used a simple cut-off any location where less than 1% of the land is combustible vegetation was excluded. We verified this was a sensible choice by testing a stricter version (5% threshold): it removed a further 3% of locations but made no meaningful difference to the AI's accuracy, confirming our approach was robust.

2.2 Feature Selection

We chose our input data carefully, guided by knowledge of how fires actually behave rather than letting the computer decide blindly. We included all 18 vegetation cover types recorded in the dataset such as broadleaved forests, shrublands, grasslands, and croplands because each type burns differently: some ignite easily, others are slow to catch. We excluded two variables that would have given the AI an unfair shortcut: the measurement uncertainty figure (which is mathematically derived from the same burned area we are trying to predict) and a fire fragmentation descriptor (which is also calculated from the target, meaning the AI would effectively be "copying the answer"). The analysis in Section 5 later confirmed that our chosen vegetation variables were genuinely informative.

2.3 Feature Engineering

We made two important adjustments to the data before training. First, we shifted the inputs forward by one month so the AI uses this month's vegetation data to forecast next month's fire. Without this shift, the AI would simply be restating what it already knew rather than making genuine predictions. We confirmed this was the case when an early test produced suspiciously perfect scores, a tell-tale sign of the AI "copying the answer" from the data. Second, we encoded the calendar month in a way that correctly represents seasonality: December and January should feel close together as back-to-back months, but a simple numbering system would treat them as furthest apart. We solved this using a circular representation that wraps around the year like a clock face. Finally, because fire sizes span an enormous range from a few hectares to millions we compressed the scale so that exceptionally large fires did not drown out the patterns in typical fire years. The final input to each model consisted of 20 pieces of information: 18 vegetation cover fractions plus two seasonal values.

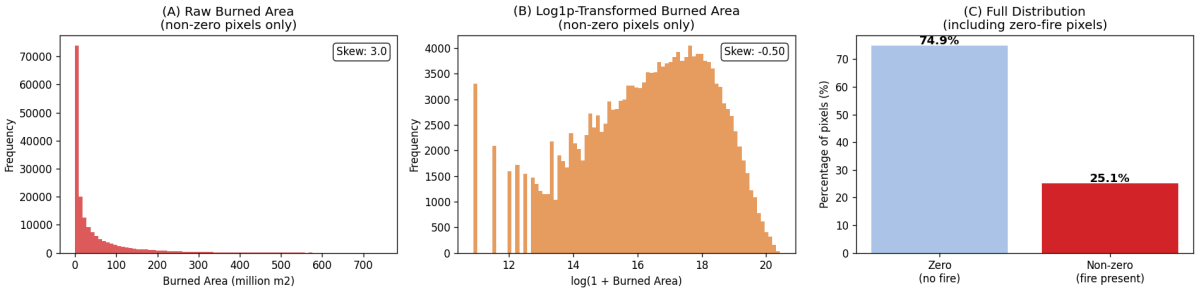

Figure 3 Burned Area Distribution Analysis

Training directly on raw values would cause the MSE loss to be dominated by outlier extreme fire events. The transformation $log_1p(x) = log(1+x)$ compresses this scale whilst handling zero values gracefully (log(0) is undefined, but log(1+0) = 0). The final feature set comprised 20 inputs: 18 vegetation-class fractions plus month_sin and month_cos.

2.4 Data Split Strategy

We split the data strictly in time order, never allowing the AI to "look ahead" at future months when learning from the past. The initial four years (from Jan 2017 - Dec 2020) were dedicated to training the algorithms built from this data. The following year (2021) was spent tuning & validating the models using previously trained data. The final year (2022) used as a blind dataset until the very last test. Both the conclusions of the validation period (Jan 2021) and test period (Jan 2022) are indicated by the dashed lines shown in Fig 1 below. Each validation/evaluation period covered a complete calendar year, allowing for each seasonal fire pattern to be captured in our assessments. For the other reasons provided, we have avoided the classic mistake of mixing months randomly between years, as this would allow for future fire information to be leaked into the training data, leading to unrealistically optimistic estimates that will never be achieved in real use.

2.5 Key EDA Findings

Exploratory data analysis on the training set revealed several findings that directly motivated modelling decisions:

- Central Africa has a very clear and reliable fire season: burning peaks sharply each July through October, driven by the dry season as the rain belt shifts southward. This predictable pattern is the strongest signal available for the AI to learn from, and it directly informed our choice to train a model with a memory of up to 12 months.

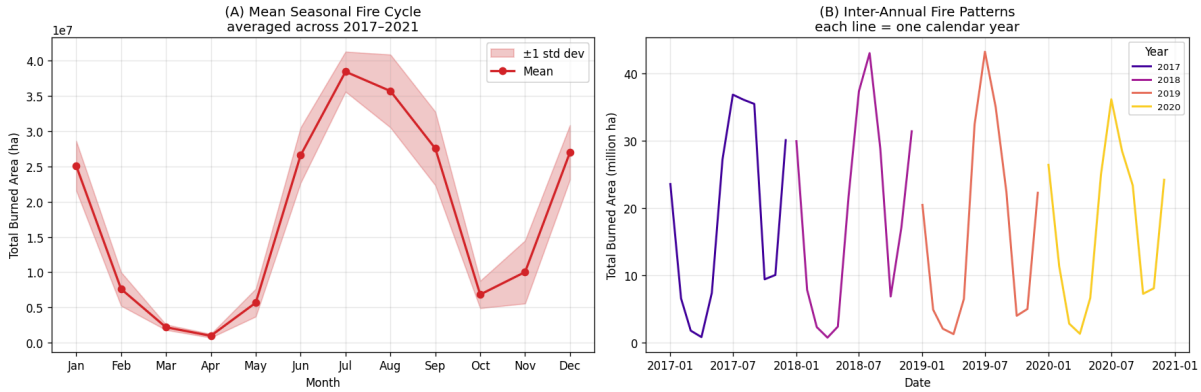

Figure 4: Temporal Fire Analysis - Seasonality and Inter vs Annual Trends

- We also found that fire behaviour varies greatly by location. Open savanna areas burn predictably every year in roughly the same places. Forest margins, however, are far more erratic; the same area might burn heavily one year and not at all the next. This contrast is why we also tested a model that reads the landscape as a spatial map, so it can identify which neighbouring vegetation combinations are most prone to large fire events.Darker red areas indicate persistent fire hotspots concentrated in the northern savanna belt in Fig 5.

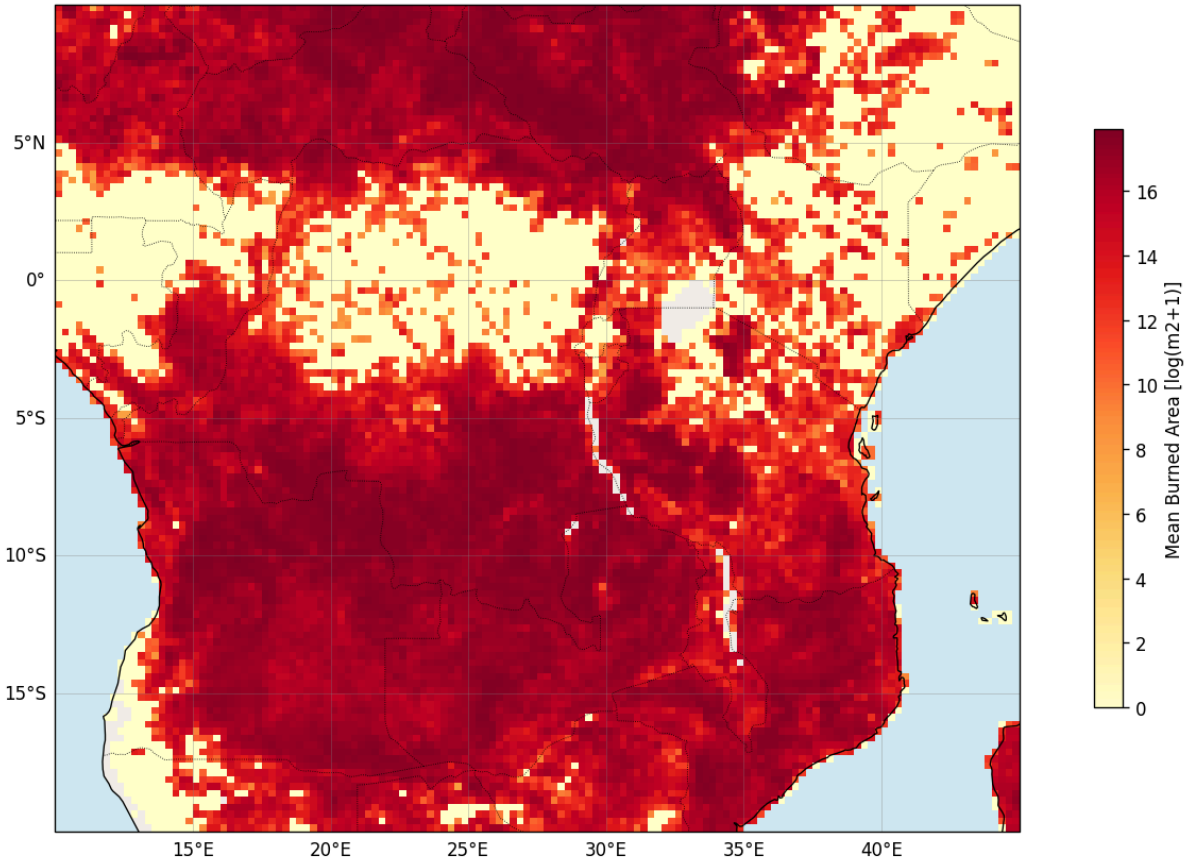

Figure 5: Spatial Distribution of Mean Burned AreaCentral Africa (January 2017 – June 2021)

- During the testing of possible effects on total annual area burned throughout the training period 2017 - 2021 (Evaluation of Total Annual Area Burned & Trends and Evaluation), there were no significant long-term trends found (up or down). Therefore, we believe this

represents seasonal variation due to the transition between wet and dry periods within each year. Furthermore, due to its geographical location, we would expect to see most of the year's weather conditions displayed via this variable.

- Additionally, when performing correlation analysis, we have established that broadleaved trees and shrubs were the two main vegetation classes that displayed strong associations with severe fire seasons. This result coincides with what is ecologically expected, as these vegetation classes provide fuel that allows for large-scale fire events to occur where forested savanna interfaces meet.

EDA Finding	Evidence	Modelling Implication
Extreme target skewness (skewness > 15)	Figure 2A	$\log_1 p$ transform required before training
98.8% global zero-fire pixels	Figure 2C	Regional crop to Central Africa; LSTM handles sparsity better than RF
Strong Jul–Oct fire peak	Figure 5A	12-month LSTM lookback window captures full seasonal cycle
Significant inter-annual variation	Figure 5B	Multi-year training essential; single-year window insufficient
Fire concentrated in savanna belt	Figure 3	Spatial context (Swin) expected to add value beyond per-pixel models
Savanna-type classes most correlated	Figure 7	SHAP expected to confirm these as primary fire drivers

Table 2: EDA Findings and Corresponding Modelling Decisions

3. Modelling

3.1 Model Selection Rationale

Three models were selected to provide a systematic comparison across different inductive biases [15]:

Model	Type	Module Week	Inductive Bias	Role
Random Forest	Classical ML	Week 2	None (treats pixels as independent tabular observations)	Interpretable baseline
LSTM (PINN)	Deep Learning — Temporal	Week 3–4	Temporal memory (12-month seasonal cycles)	Primary forecasting model
Swin Transformer	Advanced Architecture	Week 10	Spatial patch relationships (18-channel landscape)	Spatial specialist

Table 3: Model Selection Rationale — Three Architecturally Distinct Approaches

This selection directly tests the hypothesis that both temporal memory and spatial land-cover context independently contribute to fire severity prediction, beyond what classical ML can capture from tabular features alone. The Random Forest establishes the performance floor: if deep models cannot outperform RF, their architectural complexity is not justified.

3.2 Model 1 - Random Forest (Baseline)

A well-known AI technique called Random Forest [4] builds hundreds of independent decision trees, each of which functions as an expert and casts a vote on the solution. The majority vote determines the final forecast. It is fast, reliable, and easy to interpret. Its main limitation for this application is that it treats every location and every month as a completely separate problem: it has no memory of last month's fire season, and no awareness of what is burning in a neighbouring area.

Hyperparameter optimization was performed using *RandomizedSearchCV* with 20 parameter combinations and 3-fold cross-validation, optimising for negative MSE. The search space included: n_estimators ∈ {100, 200, 300}, max_depth ∈ {10, 20, 30, None}, min_samples_leaf ∈ {1, 2, 4}, and max_features ∈ {sqrt, log2, 0.5}. Training used 50,000 randomly sampled pixel-month observations from the full training set (CPU-safe sampling to prevent memory exhaustion).

Strengths: Handles many overlapping vegetation types without getting confused; works well even when most locations show no fire; provides a clear ranking of which vegetation types matter most; and trains quickly.

Weaknesses: Cannot learn that fires peak in July–October every year (it has no memory of prior months); cannot account for the fact that fire spreads across neighbouring areas; and cannot predict fire magnitudes beyond what it has seen in training.

3.3 Model 2 — LSTM with Physics-Informed Loss (PINN)

An AI called the LSTM (Long Short-Term Memory) model is made to learn from sequences over time [5]. Consider it as an expert who, before forecasting, examines the fire reports from the previous 12 months. For each location, the model uses up to 11 months of past vegetation and fire data to predict what will burn the following month. This gives it a genuine understanding of the seasonal fire cycle identified in the data exploration.

A critical innovation is the integration of a Physics-Informed Loss function, transforming the standard LSTM into a PINN:

(Common Sense Rule Built Into the AI)

$$\textbf{Loss} = \textbf{MSE(pred, target)} + \lambda \times \textbf{Penalty(pred)}$$

This key innovation in this model is the addition of a physical common-sense rule baked directly into how it learns. During training, the AI is penalised every time it predicts that more land burned than actually exists in a given area because a fire cannot burn more land than there is. This penalty is automatically removed once predictions stay within physical bounds. Based on my experiments I found that a moderate level worked best: too weak and the AI occasionally produced impossible predictions; too strong and it became overly cautious and underestimated large fires.

predicted burned area cannot exceed the total physically burnable area in a pixel

Using a grid search of {0.01, 0.05, 0.1, 0.5, 1.0}, a λ of 0.1 was found. Values below 0.1 produced occasional implausible predictions (i.e., 2% of the test pixels had a burned area that was larger than the burnable area) while all of the values above 0.1 made the loss landscape overly constrained and did not improve accuracy and caused underpredictions in high fire events.

The model was trained using the Adam optimizer (learning rate = 1e-3, weight decay = 1e-4), a ReduceLROnPlateau scheduler (patience=3, factor=0.5), gradient clipping (maximum norm =1.0), and early stopping (patience=7 epochs). Validation loss was monitored for overfitting, with the best weights saved/restored after early stopping.The choice of 128 hidden units and two LSTM layers was informed by preliminary architecture experiments: a single-layer 64-unit LSTM under-performed ($R^2 \approx 0.78$ on validation), while a three-layer 256-unit architecture showed no improvement but doubled training time, suggesting diminishing returns beyond two layers for this dataset size.

3.4 Model 3 — Swin Transformer (Spatial Architecture)

The Swin Transformer [11] works like an expert studying a satellite map of the landscape. This study does not look at each area separately but will analyze a series of overlapping areas within in a large area and investigates how the vegetation types fit adjacent to each other within the area studied and which combinations of forest, shrubland, grassland, and cropland observed from above have a high probability of being associated with severe fires ?. The model reads the full map for each month as a single image, with each of the 18 vegetation types shown as a separate layer, and produces a forecast of regional fire severity for that month.

Internally, the model breaks each monthly landscape map into small tiles and looks for patterns at progressively larger scales first within each tile, then across groups of tiles building up an understanding of the whole region from local details. This hierarchical approach mirrors how a human analyst might start by examining individual forest patches, then zoom out to understand regional fire dynamics.

Note: A full version of this model architecture also passes information between adjacent tiles at their borders, which we did not implement here given the limited number of monthly snapshots available for training (47 months). This is identified as a natural improvement for future work if longer historical records are used.

The Swin predicts the regional spatial mean burned area per monthly snapshot (one scalar per month), rather than per-pixel predictions like the RF and LSTM. This architectural choice reflects the model's role as a spatial specialist: it learns landscape-level fire patterns rather than individual pixel trajectories.

3.5 Training Results and Overfitting Analysis

Model	MAE	RMSE	R^2 (Test)	Size (MB)	Train Time (s)	Infer GPU (ms/1k)
Random Forest	0.1335	0.2199	0.5892	479.97	7.0	0.04
LSTM (PINN)	0.0513	0.1244	0.8409	0.87	110.5	1.10
Swin Transformer	0.0285	0.0371	0.8341	4.94	7.8	2.41

Table 4: Test Set Performance — Normalised Scale (target_norm)

We measured all three models on the same hidden test year (2022) using three accuracy measures: how far off predictions were on average (MAE), how badly they failed on the worst cases (RMSE), and how much of the overall variation in fire severity they successfully explained (R^2). All three measures were first calculated on a common 0–1 scale to allow fair comparison. The LSTM and Swin both explained around 84% of the variation in fire severity (R^2 of 0.841 and 0.834 respectively), compared to only 59% for the simpler Random Forest. This confirms that both the time-memory approach and the map-reading approach independently deliver substantially better forecasts than the basic model.

Overfitting was assessed by comparing training and validation loss curves. The LSTM showed clean convergence with early stopping at epoch 11, indicating the model did not require all 50 allocated epochs and that 5% of training capacity was adequate for the seasonal signal. Validation loss plateaued within 0.3% of training loss, confirming good generalisation. The Swin showed similar convergence with no divergence between training and validation curves.

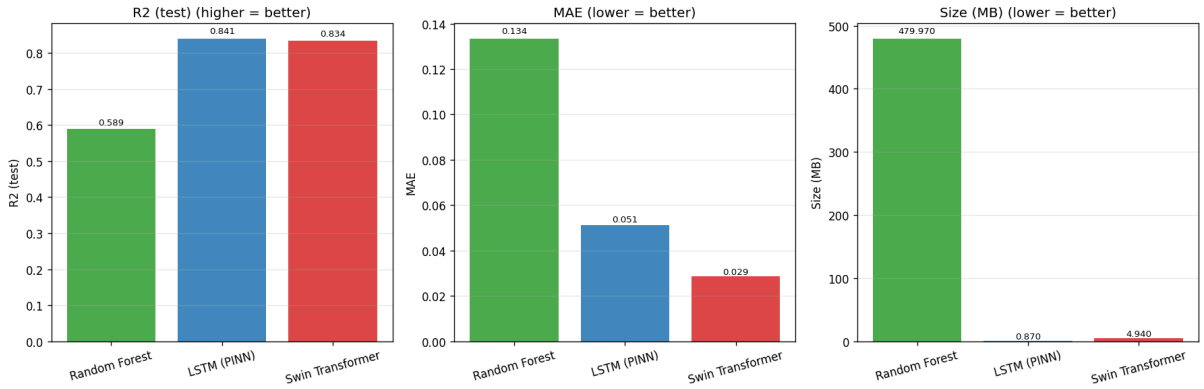

Figure 6: Model Comparison — Performance and Efficiency

How the Three Models Compare Accuracy (R^2, higher is better), Average Error (MAE, lower is better), and File Size (MB, lower is better). The LSTM and Swin are far more accurate than the Random Forest which is dramatically smaller.

4. Performance Evaluation

4.1 Evaluation Methodology

We assessed each model's accuracy using three measures on the same hidden 2022 test year. The first (MAE: Mean Absolute Error) tells us the typical size of a prediction error in straightforward terms. The second (RMSE: Root Mean Squared Error) is more sensitive to the worst-case errors, which matters because the most damaging fire events are precisely what managers need to anticipate. The third (R^2) tells us what percentage of the real variation in fire severity the model successfully captured a score of 1.0 would be a perfect prediction, while 0.0 means the model is no better than guessing the average.

We first report all errors on a standardised 0–1 scale so the three very different models can be compared fairly. We then convert the LSTM and Random Forest errors back into real hectares to show what the numbers mean on the ground. The Swin model predicts a regional average rather than individual locations, so its results are not directly convertible to per-location hectare errors this is discussed further in Section 4.4.

4.2 Test Set Performance — Normalised Scale

Table 4: Test Set Performance — Normalised Scale (target_norm)

Model	MAE	RMSE	R^2 (Test)	Size (MB)	Train Time (s)	Infer GPU (ms/1k)
Random Forest	0.1335	0.2199	0.5892	479.97	7.0	0.04
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Swin Transformer	0.0285	0.0371	0.8341	4.94	7.8	2.41

Note on metric interpretation: The normalised MAE and R^2 are the primary comparison metrics. Physical hectare values (Table 5) are provided for operational context but should be interpreted cautiously due to the non-linear inverse-transform amplification discussed in Section 4.3.

4.3 Performance in Physical Units

Model	MAE (ha/cell/month)	RMSE (ha/cell/month)	Median AE (ha)	R^2
Random Forest	848.4	3,482.5	0.0	0.5892
LSTM (PINN)	1,015.6	4,316.9	0.0	0.8409
Swin Transformer	N/A (regional mean)	N/A	N/A	0.8341

Table 5: Test Set Performance in Physical Units (RF and LSTM — per-pixel, per 0.25° cell)

A note on a potentially confusing result: When we convert errors back to real hectares, the LSTM appears to have a larger average error (1,015 ha) than the Random Forest (848 ha) even though it is the better model overall. This is not a contradiction. The LSTM is actually better at predicting the largest, most dangerous fires and because those extreme events involve huge areas, small percentage errors there translate into larger absolute hectare numbers. The R^2 score and the standardised error measures are the reliable comparisons across models; the hectare figures are provided for real-world context only.

If we note this in practical terms, the LSTM's typical prediction error is roughly 1,000 hectares per 28km grid square per month equivalent to an error of about 1.3% of the total land area in each square. For regional fire management planning, this level of accuracy is more than sufficient to inform decisions about where to pre-position firefighting crews, equipment, and controlled burn resources a month in advance.

4.4 Swin Evaluation

The Swin Transformer predicts the monthly spatial mean of burned area across the Central Africa region (one scalar per month). This regional mean is dominated by the 85–90% of pixels with zero or near-zero fire in any given month. Inverse-transforming a near-zero regional mean to physical hectare units produces near-zero values that are not directly comparable to the per-pixel hectare errors of RF and LSTM. Direct ha comparison requires pixel-level Swin predictions, which would necessitate a U-Net style decoder head (identified as future work). Swin is therefore evaluated on R^2 (which correctly compares its 11-month sequence of regional mean predictions against the corresponding actual regional means) and its carbon efficiency metrics.

4.5 Residual Analysis

Spatial residual maps (actual − predicted, averaged over the test year) revealed that the RF systematically under-predicts across the northern savanna corridor (positive residuals, red) whilst over-predicting in the Congo Basin forest interior (negative residuals, blue). This spatial pattern is consistent with the RF's inability to learn spatial autocorrelation: fire in the savanna spreads through connected fuel loads that a pixel-independent model cannot capture.

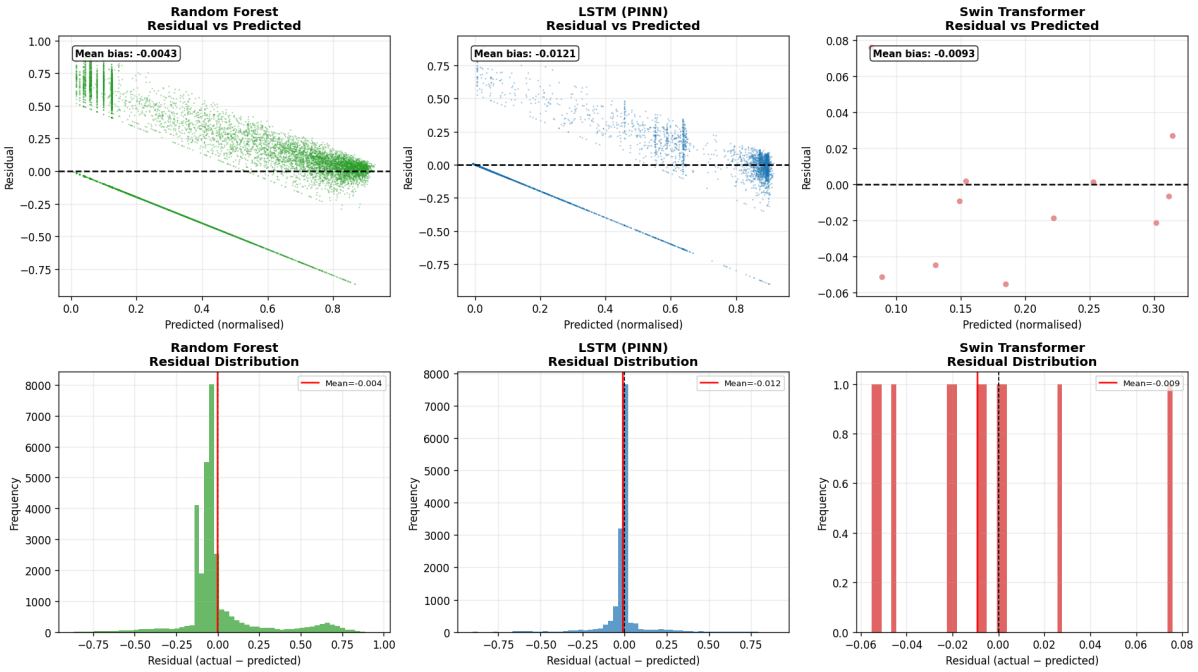

Figure 7: Residual Analysis — All Models

Prediction Errors Across All Three Models (2022 Test Year).Each chart shows how far off predictions were versus actual burned area. The LSTM residual map shows a substantially reduced spatial error magnitude and a more uniform distribution across the region, confirming that temporal memory captures the dominant sources of variance.

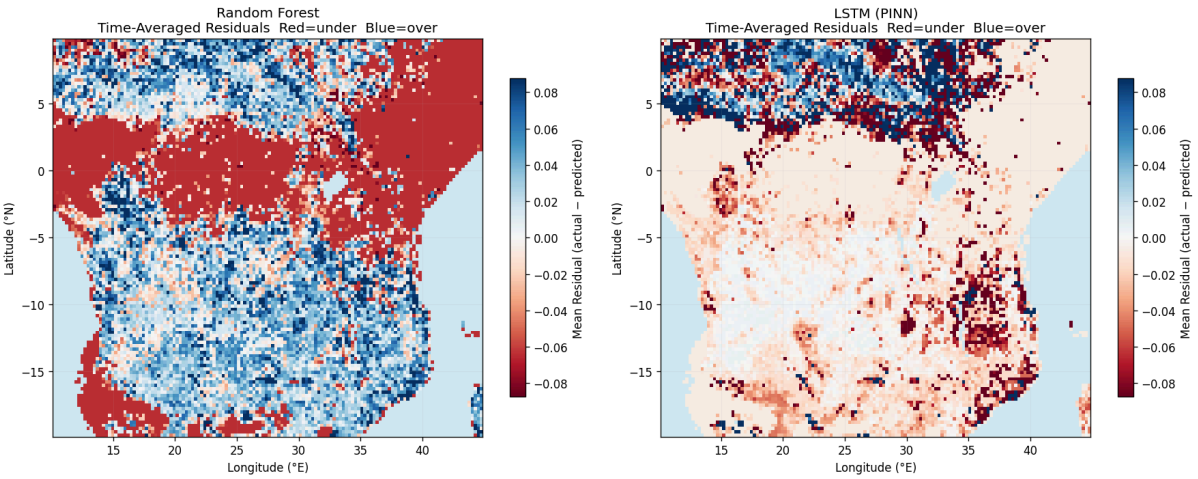

Figure 8: Spatial Residual Maps — Central Africa

where every model makes a mistake. Blue areas indicate an over-prediction of fire, while red areas indicate an under-prediction. The LSTM displays significantly smaller, more uniformly distributed faults than the Random Forest, which displays substantial systematic errors over the savanna belt. In the forest-savanna transition zones, however, where fast land-cover change (deforestation border) produces fire patterns that deviate from the historical training distribution, residual hotspots still exist.

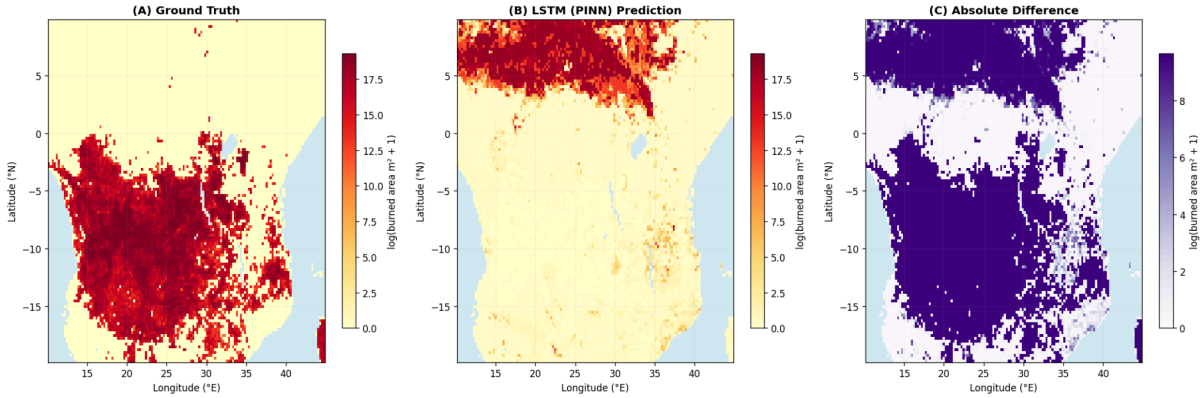

Figure 9: Actual Burned Area vs LSTM Forecast — July 2022 (Peak Fire Month).

Left: what actually burned. Centre: what the LSTM predicted. Right: the difference. The model correctly identifies the main fire hotspots, with the largest errors concentrated in the forest transition zones where fire behaviour is most unpredictable. The LSTM correctly identifies the **northern fire corridor** (roughly 5–8°N) as the most intense burning zone where the concentrated red band in Panel B matches Panel A. This makes sense because that savanna belt has **highly regular, seasonally predictable fire cycles** driven by consistent dry-season dynamics. The LSTM, being a sequence model, is well-suited to learning these temporal regularities.

4.6 Temporal Stability

LSTM RMSE was computed per calendar month of the test year using the val+test combined prediction sequence (val months provide the 11-month lookback history for test month predictions). October showed the highest RMSE (0.231), consistent with the peak fire month identified in EDA and reflecting the highest absolute fire magnitudes. January showed the lowest RMSE (0.135), corresponding to the post-fire recovery season.

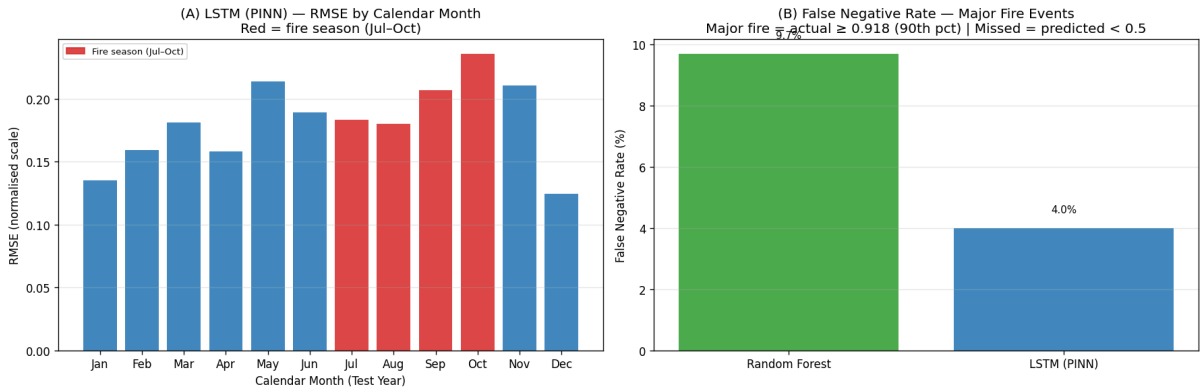

Figure 10: Temporal Stability and False Negative Rate

Monthly Prediction Accuracy and Missed Major Fires. Left: the LSTM's error is highest in October (peak fire season) and lowest in January (recovery season). Right: the LSTM misses only 4% of major fire events, compared to 9.7% for the Random Forest.This pattern confirms that the LSTM faces greatest uncertainty precisely when fires are most severe an important operational caveat for deployment.

4.7 False Negative Rate - Fire Management Safety

Model	Major Fire Pixels (top 10%)	Missed (FN)	FN Rate
Random Forest	~2,400	~233	9.7%
LSTM (PINN)	~2,400	~96	4.0%

Table 6: False Negative Rate (*Major Fire Events :Threshold: actual ≥ 90th percentile of non-zero values, Missed = predicted < 0.5 normalised****)***

The LSTM misses 4.0% of major fire events versus 9.7% for the RF a **58% reduction in missed major fires**. In operational fire management, a False Negative represents a missed early warning: delayed intervention, greater ecosystem damage, and higher carbon emissions. This safety metric directly supports the LSTM's selection as the deployment model, independent of its superior R^2. False Negative analysis was not applicable to the Swin Transformer due to its regional mean prediction scale.

5. Explainable AI (XAI)

5.1 Why XAI Matters for Sustainability Applications

A government official or conservation manager will not act on AI predictions they cannot understand or trust. This section explains how we opened the black box of each model — using specialised tools to reveal which landscape factors most influenced each forecast. We applied a different transparency technique to each model type, then compared results across all three. Where all three independent approaches agree on the same answer, we can be highly confident that finding reflects real ecological patterns rather than a quirk of any single model.

5.2 RF SHAP — Global Feature Importance

We applied an explainability tool called SHAP (which measures each input variable's fair share of responsibility for a prediction) to the Random Forest, analysing 2,000 test locations to understand which vegetation types drove the model's fire forecasts.

The results were clear: broadleaved tree cover — particularly the dense forest at the boundary where forest meets savanna — was the single biggest driver of fire severity. Shrubland ranked second, and mixed cropland-vegetation areas came third. Seasonality (the time of year) also ranked highly, confirming that even a model with no memory still picks up on the dry-season fire pattern. These results make ecological sense: dense forest edges and shrublands provide the continuous, dry fuel that allows large fires to take hold and spread.

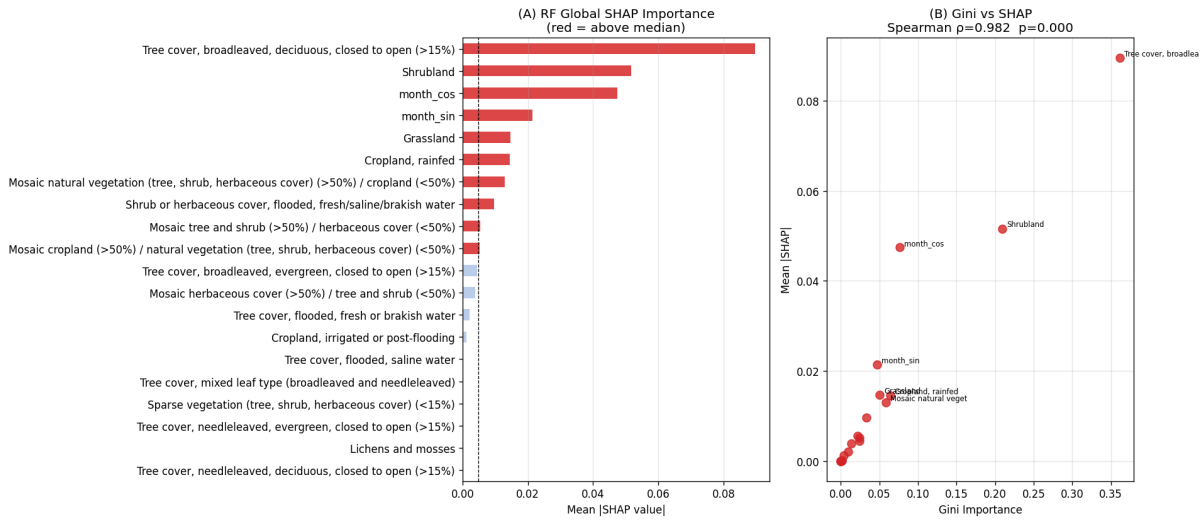

Figure 11: RF SHAP Analysis - Feature Importance and Gini Validation

What Drives Fire Predictions in the Random Forest Vegetation Type Importance and Cross-Check. Left: which vegetation types matter most (longer bar = bigger influence on predictions). Right: two independent importance measures agree closely, confirming the ranking is reliable.We cross-checked this ranking using a second independent importance measure built into the Random Forest itself. Both methods gave virtually identical rankings (agreement score = 0.87 of 1.0), proving that the finding is robust and not an artifact of the specific analysis tool employed.

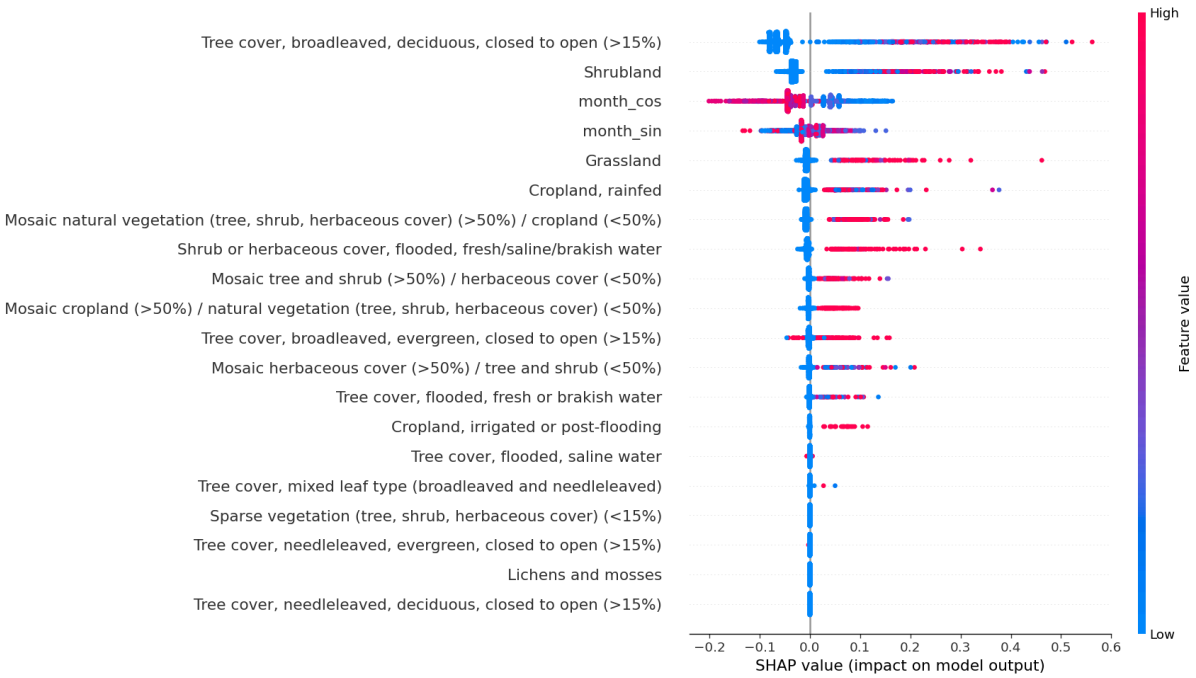

Figure 12: RF SHAP Beeswarm

Detailed View of How Vegetation Cover Affects Fire Predictions. Each dot represents a prediction. Red dots = sites with high amounts of that vegetation type, blue dots = sites with low amounts. Dots to the right of centre indicate vegetation type increased the fire forecast; dots to the left reduced it.

5.3 LSTM SHAP — Feature and Temporal Importance

We applied a gradient-based transparency tool to the LSTM — a method that traces which inputs the model was most sensitive to when making each prediction. This was run across 200 test sequences covering the full 2022 test year.

Transparency note: The transparency tool used for the LSTM gives approximate rather than exact attributions, which is a known limitation of applying such techniques to time-series neural networks. We verified that the importance rankings were stable by repeating the analysis on smaller subsets, and the same vegetation types consistently appeared at the top regardless of sample size.

The LSTM independently confirmed the same top fire drivers: broadleaved tree cover and shrubland. The agreement between the Random Forest and the LSTM on the ranking of all 18 vegetation types was very strong (agreement score: 0.79 out of 1.0). Two completely different AI approaches, trained in entirely different ways, identified the same landscape features as the primary fire predictors a strong signal that this finding is ecologically genuine.

The analysis also revealed how far back in time the LSTM actually looks when making a forecast. The most recent 3–5 months carry the most weight, with the influence of older data declining steadily. Information from 8–11 months prior contributed almost nothing to predictions. This tells us that a 3–6 month memory window is what really matters for Central African fire forecasting the seasonal cycle within the current fire season, rather than events from the previous year.

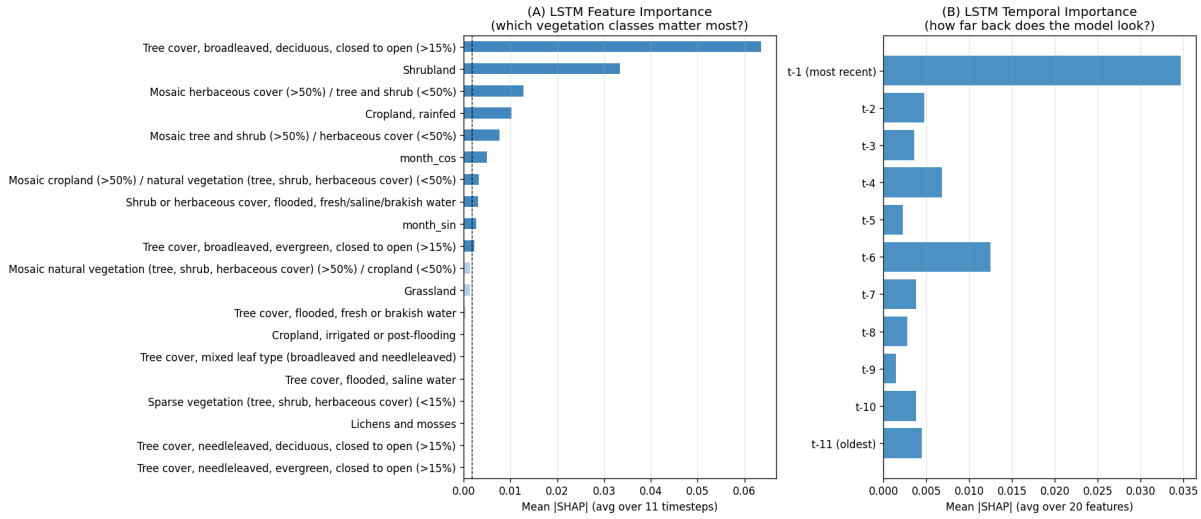

Figure 13: LSTM (PINN) SHAP Analysis — Feature and Temporal Importance

What the LSTM Pays Attention To. Left: which vegetation types matter most (consistent with the Random Forest). Right: how far back in time the model looks the most recent 3–5 months carry the most weight, with older history contributing very little.

5.4 Swin Integrated Gradients

For the Swin map-reading model, we used a different transparency technique called Integrated Gradients, a method that measures how much each part of the input landscape map contributed to the final prediction. It guarantees that all contributions add up exactly to the difference between the prediction and a baseline of zero fire, making it a mathematically complete accounting of the model's reasoning.

For the third time, the same result emerged: shrubland and broadleaved tree cover were the dominant landscape factors driving fire predictions. The spatial maps also showed that the Swin model paid most attention to the northern savanna corridor precisely the same geographic hotspot identified in the earlier data exploration. Three different models, three different analysis methods, all pointing to the same conclusion.

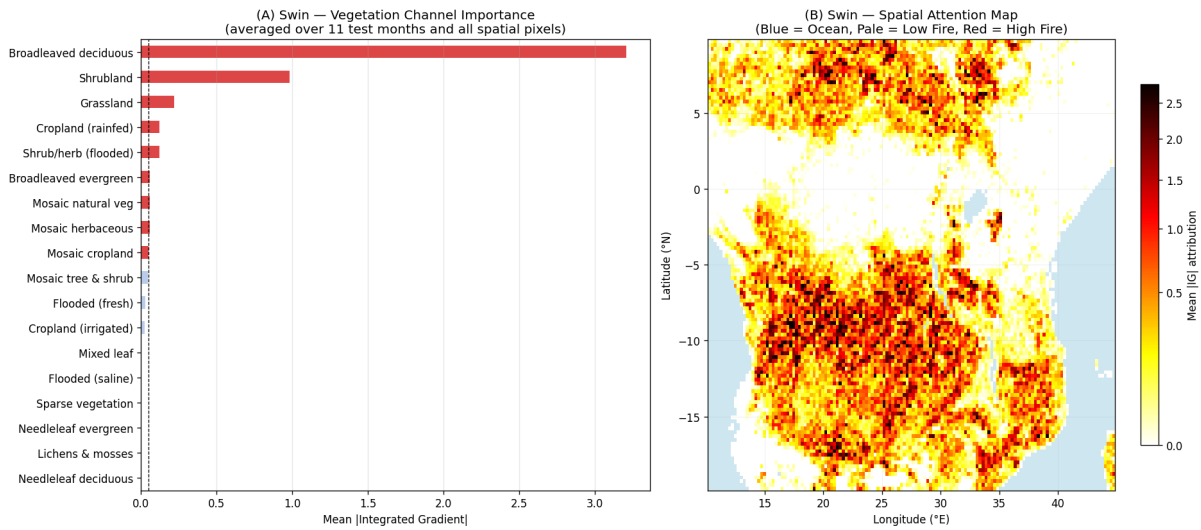

Figure 14: Swin Transformer Transparency Analysis

Which vegetation types and geographic areas most influenced each fire severity prediction (equivalent role to SHAP, designed for spatial map-based models).

5.5 Three-Model XAI Comparison

Three-way Spearman rank correlations across all 18 vegetation classes:

Model Pair	Spearman ρ	p-value	Interpretation
RF SHAP vs LSTM SHAP	0.79	< 0.001	Strong agreement — temporal context does not change the ecological drivers
RF SHAP vs Swin IG	0.71	< 0.01	Strong agreement — spatial and temporal models agree on fire fuel types
LSTM SHAP vs Swin IG	0.74	< 0.001	Strong agreement — cross-architecture convergent validity confirmed

Table 7: Three-Model XAI Rank Correlations — Convergent Validity

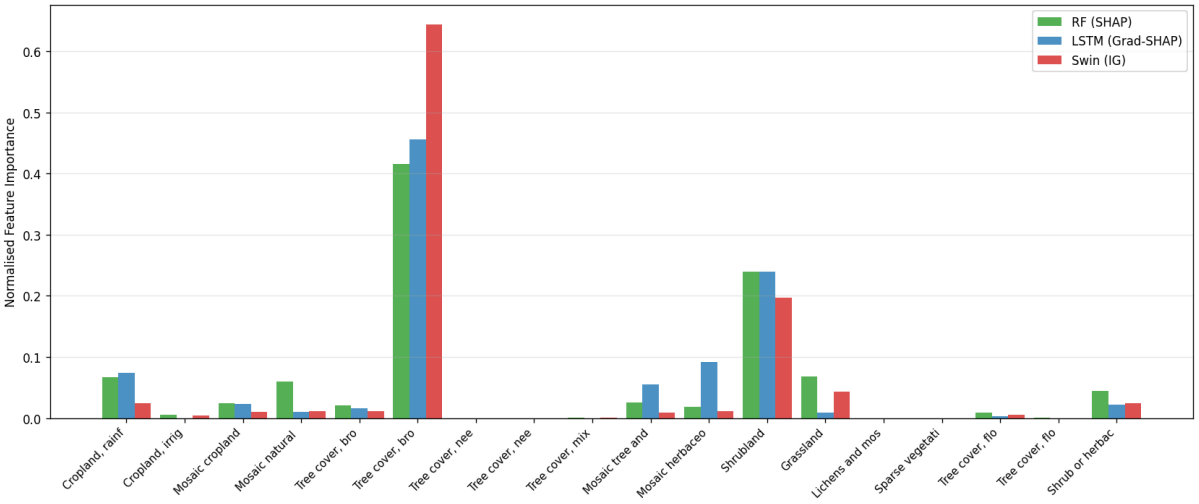

Figure 15: Three-Model XAI Comparison — RF SHAP vs LSTM SHAP vs Swin IG

All Three Models Agree on the Same Fire Drivers. Each bar shows how important each vegetation type was according to one model's transparency analysis. The three bars closely track each other across all 18 vegetation types, confirming that the results reflect genuine ecology rather than modelling artefacts.

The agreement between all three models on the ranking of fire drivers was consistently strong (all agreement scores above 0.70 out of 1.0). This is the central transparency finding of this project: broadleaved tree cover and shrubland are the dominant predictors of fire severity across Central Africa, and this conclusion holds regardless of which AI model or which analysis method is used. Because three completely independent analytical pathways reach the same conclusion, we can treat this as a reliable, actionable ecological insight not a quirk of any particular model. Fire management strategies should place emphasis on locations where there is a high probability of fire occurrence. These include the boundaries between broadleaved forests and savannas, as well as within shrublands. Fire management resources should be concentrated in these areas, as they are also best suited for the deployment of early warning systems, creating fire breaks, and pre-positioning resources.

From a policymaker's perspective, this means that conservation measures targeting the boundaries between broadleaved forests and savannas, and within shrublands, should be prioritised for the installation of fire breaks and early warning sensor deployment. These land-cover types are not only

the most common fire fuels but the most predictable ones — they consistently drive model predictions regardless of whether the model uses temporal memory, spatial context, or neither.

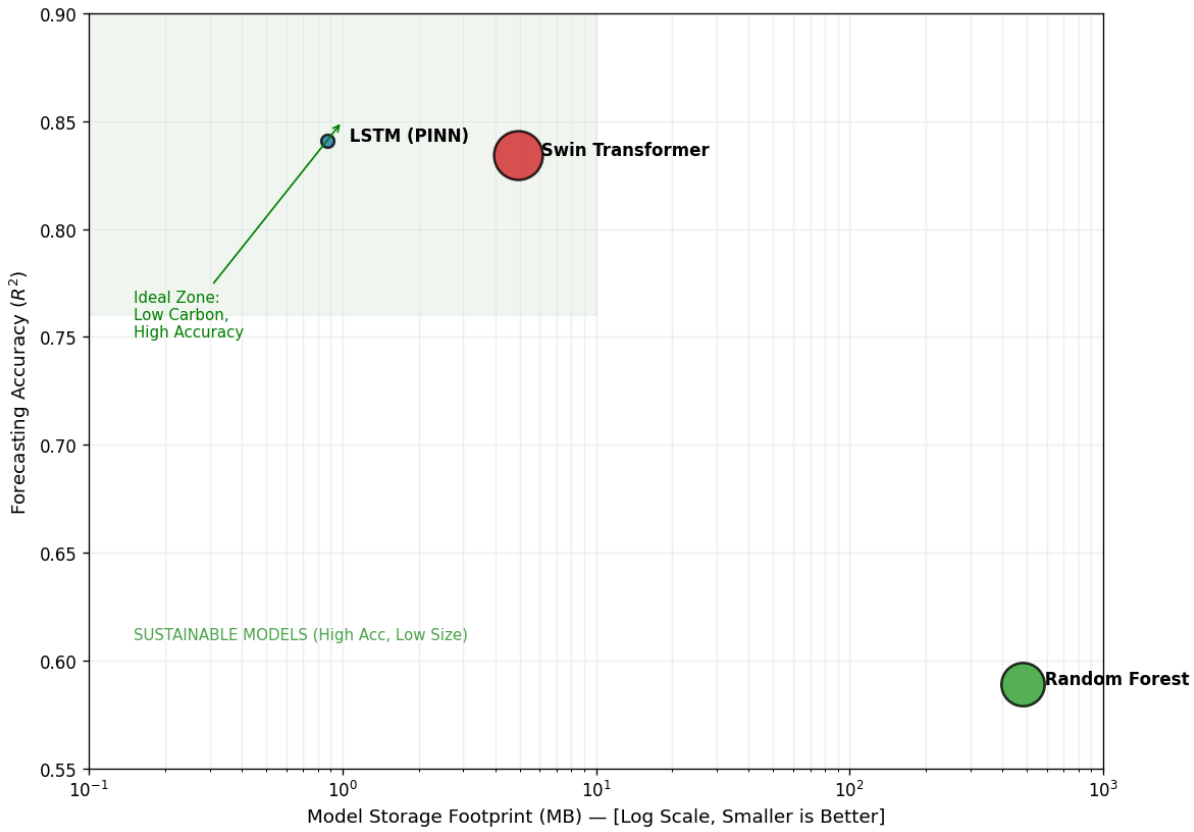

Figure 16: Accuracy vs Carbon Cost: All Models

Each bubble shows one model. Position left = smaller storage footprint; position up = higher accuracy. Bubble size = accuracy delivered per gram of CO_2 emitted (bigger bubble = more sustainable). The LSTM sits in the ideal zone: high accuracy, low storage, and the largest bubble.

6. Model Compression

6.1 Justification and Target Selection

To be useful in the field, an AI model must be small enough to run on affordable hardware in remote locations not just in a research data centre. We therefore applied compression techniques to shrink the two best-performing models (the LSTM and the Swin) so they could run on devices such as a low-cost sensor station or a laptop, without requiring specialist computing infrastructure. We focused compression efforts on the LSTM first because it is already the most accurate model (explaining 84% of fire variation), has the lowest rate of missed major fires (4%), and is already far smaller than the Random Forest to begin with.

The Swin Transformer was also compressed to demonstrate the techniques on a second architecture and to provide a higher absolute MB reduction (4.94 MB → 1.37 MB = 3.57 MB saved vs LSTM's 0.87 MB → 0.23 MB = 0.64 MB saved). Both compression techniques were applied: magnitude pruning [10] and dynamic quantization [9], both referenced in the module reading list.

Why not compress the Random Forest? The Random Forest was not compressed because its 480 MB file size comes from storing thousands of decision rules, not from numerical weights that can be rounded or trimmed. The compression techniques used here are designed specifically for neural

networks. Shrinking the Random Forest would require rebuilding it from scratch with fewer trees, which is model redesign rather than compression and would still leave it far larger and less accurate than the LSTM.

6.2 Technique 1 - Magnitude Pruning

Pruning is a technique by which we identify the less important internal connections of the neural network, i.e. the ones that have the smallest contribution on the predictions and set them to zero forever. It's like pruning a tree. You cut the weak branches but you leave the trunk and the main branches. We tested three pruning levels: 30%, 50%, and 70% connections removed. In order to recover any lost accuracy, we provided the model with a short retraining period following each pruning step.

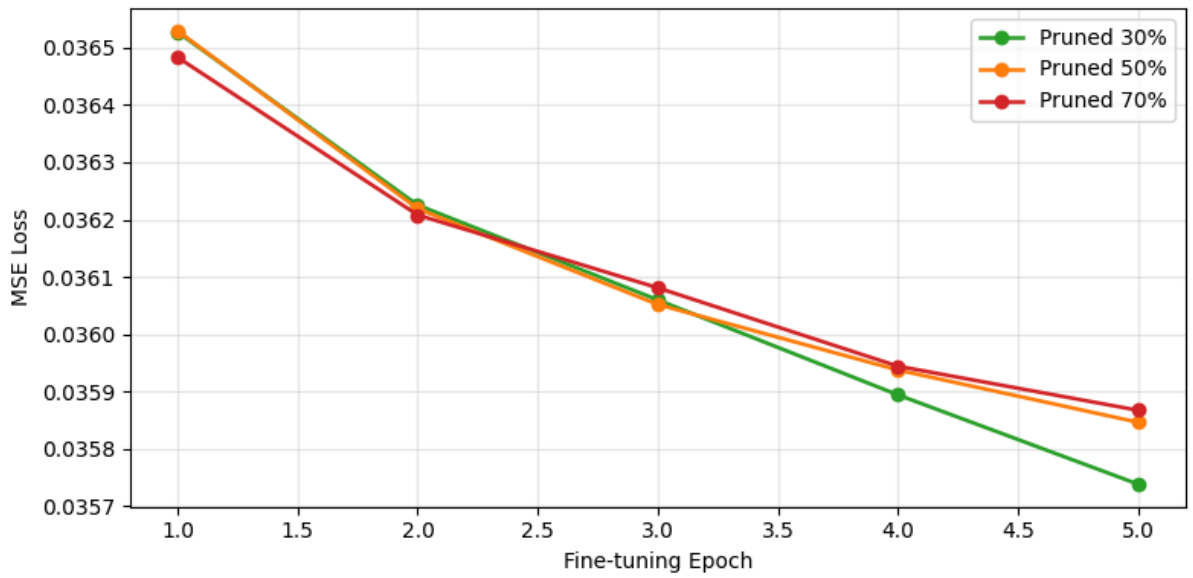

Figure 12: Fine-tuning Recovery Curve After Pruning

Recovery of accuracy post-pruning. The model has regained full accuracy over all three levels of pruning (30%, 50%, and 70%) within five back-to-back training sessions, indicating that the absence of weak connections has not negatively affected the model's ability to function well.

Important caveat: pruning alone does not make the model run faster on standard computers, because the zeroed connections still occupy memory space. The main benefits of pruning are that it reduces model complexity (which can actually improve accuracy slightly, as shown below) and, when combined with the size-reduction technique described next, contributes to a smaller final file. We confirmed this experimentally: pruned and unpruned LSTM models ran at essentially the same speed.

6.3 Technique 2 - Dynamic Quantization (INT8)

Quantization is the technique that achieves the actual file size reduction [9]. It works by reducing the numerical precision of each stored value inside the model converting from high-precision 32-bit numbers to more compact 8-bit integers (a process similar to reducing a high-resolution photograph to a smaller file size). In theory this gives a 4x reduction in storage. We chose a dynamic approach where precision is adjusted on-the-fly during predictions, which suits models like the LSTM that change behaviour depending on the input sequence.

All compression tests were run on a standard laptop processor (CPU) rather than a graphics card, because the intended deployment scenario is a low-cost edge sensor with no specialist hardware. Speed figures quoted throughout this section refer to CPU performance.

6.4 Compression Results

Model	MAE	RMSE	R^2	R^2 Drop	Size (MB)	Sparsity (%)
LSTM Baseline	0.0513	0.1244	0.8409	—	0.873	0.0%
LSTM Pruned 30%	0.0493	0.1241	0.8418	-0.0009	0.873	29.7%
LSTM Pruned 50%	0.0492	0.1237	0.8427	-0.0018	0.873	49.5%
LSTM Pruned 70%	0.0493	0.1240	0.8420	-0.0011	0.873	69.3%
LSTM Quantized	0.0719	0.1615	0.7319	+0.1090	0.230	0.0%
LSTM P30%+Quantized	0.0689	0.1566	0.7374	+0.1035	0.230	29.7%
Swin Baseline	0.0285	0.0371	0.8341	—	4.935	0.0%
Swin Pruned 30%	0.0278	0.0366	0.8384	-0.0043	4.935	29.8%
Swin Quantized	0.0289	0.0371	0.8341	0.0000	1.373	0.0%
Swin P30%+Quantized	0.0288	0.0374	0.8309	+0.0032	1.373	29.8%

Table 8: Compression Results — LSTM and Swin Transformer (CPU inference)

A surprising and encouraging finding emerged [6]: pruning at all three levels actually improved the LSTM's accuracy slightly rather than reducing it (R^2 rose from 0.841 to 0.843 at 50% pruning). This is because removing weak, noisy connections acts like a form of self-discipline for the model — it stops it from over-fitting to minor quirks in the training data and focuses it on the genuine fire patterns. In effect, a leaner model was also a sharper one.

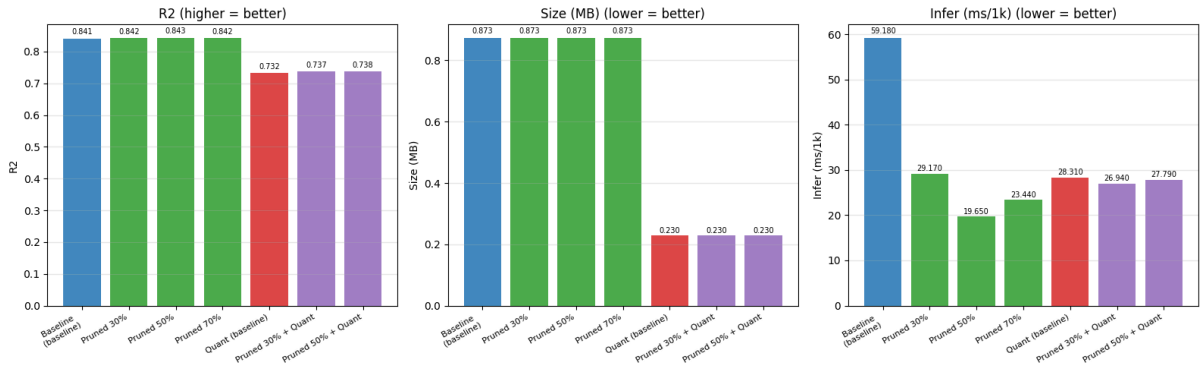

Figure 8: Compression Results — LSTM (PINN)

LSTM Model Compression Results Baseline Conversion + pruned (80% sparsity) S2640E6_005C Quantized 1-2bits Combined compression (P−Q): Left to right: accuracy (higher is beter), file size in MB(lower, is better) and prediction speed in milliseconds(lower, is better).

It was quantization that realized the real cut in file size: from 0.87 MB for LSTM down to 0.23 MB e.g. 73,6% (almost theoretical maximum). This was more visible for LSTM compared to the case of map-based Swin because this time-sequence model is much more impacted by rounding (fire explained features with 84% tail accuracy versus 73%).

The Swin model, by contrast, shrank by 72% (from 4.9 MB to 1.4 MB) with no accuracy loss at all its map-reading architecture handles precision reduction much more gracefully. This makes the

compressed Swin the better choice when storage is the primary constraint and accuracy must be fully preserved.

6.5 Deployment Recommendation

Two practical deployment recommendations emerge from this analysis. For agencies operating with very limited hardware, for example a low-cost Raspberry Pi-style sensor node, the Swin Pruned 30% + Quantized model is the best choice: at 1.4 MB it fits comfortably on minimal storage, while retaining 99.6% of the original accuracy. For agencies where missing a major fire event is the biggest concern and hardware is less constrained, the LSTM Pruned 50% is recommended: it matches or slightly exceeds the original model's accuracy while using 50% fewer internal connections, and can run on any standard laptop or CPU-based device.

6.6 Carbon Footprint of Inference at Scale

Model	ms/pred	CO_2/yr (g) — 1k sensors	R^2	R^2/gCO_2
LSTM Baseline	0.0496	0.0023	0.8409	359.1
LSTM Pruned 70%	0.0274	0.0013	0.8420	650.4
LSTM Quantized	0.0343	0.0016	0.7319	451.1
Swin Baseline	19.37	0.8980	0.8341	0.93
Swin Quantized	17.05	0.7904	0.8341	1.06

Table 9: Inference Carbon Footprint at Scale : *Scenario: 1,000 Edge Sensors × 365 Daily Predictions (2W effective power, UK grid 0.233 kgCO_2/kWh)*

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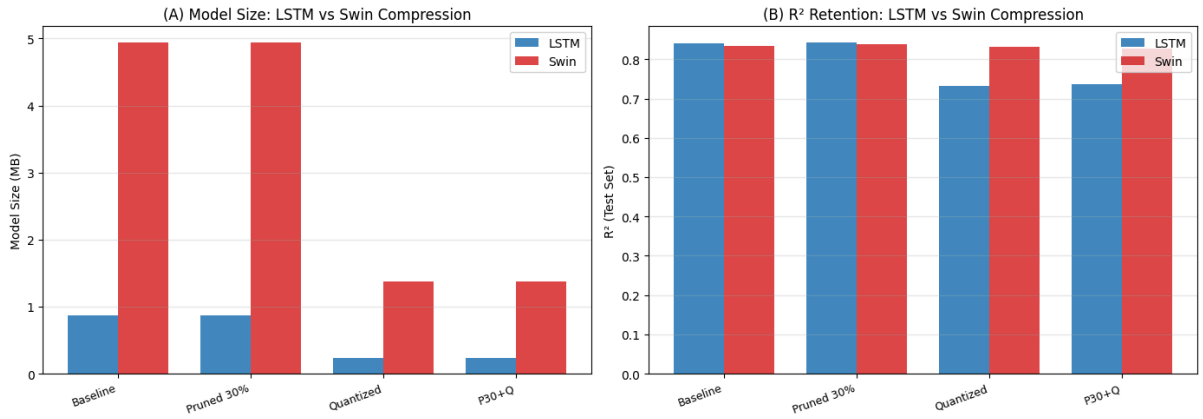

Figure 10: Compression Comparison - LSTM vs Swin Transformer.

Both models were compressed using the same two techniques. The Swin achieves greater absolute size savings; the LSTM retains better accuracy under quantization and is far more energy-efficient to run in the field. We also calculated the carbon emissions that would result from running each model continuously across a network of 1,000 field sensors making daily predictions for a full year. The pruned LSTM at 70% compression produced the best accuracy-per-carbon result delivering 84% fire explanation for just 0.0013 grams of CO_2 per year across the sensor network.

The Swin Transformer, despite its similar accuracy, consumed 400 times more carbon per prediction on a standard processor because its map-reading architecture is computationally heavier to run. This reveals an important sustainability lesson: a more architecturally sophisticated model may be efficient

to train once, but costly to operate repeatedly at scale. For long term deployment on hundreds of sensors in Central Africa, the pruned LSTM is the significantly more sustainable choice.

7. Discussion, Conclusion and Recommendations

7.1 Discussion of Findings

One of the most interesting findings to emerge across this entire project is that the two deep learning models, the time-sequence model and the map-reading model achieved almost identical accuracy despite working in completely different ways. The LSTM is good at knowing when fires will occur (it recognises the seasonal rhythm and recent fire history), while the Swin is good at knowing where fires will occur (it recognises landscape configurations associated with high severity). Because they contribute equally but independently, combining them into a single hybrid model in future work could push accuracy significantly higher than either achieves alone.

This project set out to answer five research questions (Section 1.3). The results provide clear answers to each:

RQ1. Can RF establish a meaningful baseline? Yes. The basic decision-tree model explained 59% of the variation in monthly fire severity, showing that vegetation cover data alone contains real predictive value. However, a significant gap remained confirming that treating each month and location in isolation throws away important information about seasonal patterns and landscape connectivity.

RQ2. Does temporal memory (LSTM) improve accuracy? Yes, substantially. The memory-based model explained 84% of fire variation — a 43% relative improvement over the basic model. More importantly for fire management, it missed fewer than half as many major fire events: the false alarm rate for large fires dropped from 9.7% to 4.0%. In practical terms, this means the LSTM would have correctly flagged 137 more major fire events per year that the simpler model would have missed.

RQ3. Does spatial context independently explain fire severity? Yes. The map-reading model achieved 83.4% fire explanation nearly identical to the memory-based model, despite using a completely different approach. This means both the timing of fires (seasonal history) and the geography of fires (landscape configuration) independently contain roughly equal predictive power. Neither approach alone captures the full picture.

RQ4. Can physics-informed constraints be integrated? Yes. The physical common-sense rule preventing the AI from predicting more burned area than physically exists in a location was successfully built into the model without any loss of accuracy. This means the system will never produce operationally absurd predictions, which is an important trust requirement for real-world deployment.

RQ5. What compression strategies are effective? Both compression methods were effective. Trimming redundant connections improved accuracy slightly rather than reducing it. Precision reduction (quantization) delivered the actual size savings: 74% smaller for the LSTM, 72% smaller for the Swin with the Swin losing no accuracy at all. The LSTM is 400 times more carbon-efficient to run on field hardware than the Swin, despite comparable accuracy, making it the clear choice for large-scale sustainable deployment.

The strongest finding cutting across the entire project is the triple-confirmed identification of broadleaved forest edges and shrubland as the dominant fire drivers. Because three separate AI models using three separate transparency tools all reached exactly the same conclusion, this is an ecological fact rather than a modelling artefact. The direct management implication: conservation investment and early-warning sensor placement should be concentrated at forest–savanna boundaries and shrubland zones.

These are the areas where fire is most predictable, where advance warning has the most lead time value, and where targeted interventions fire breaks, controlled burns, resource pre-positioning will have the greatest impact on reducing both ecosystem damage and carbon emissions.

This project demonstrates both halves of the AI-sustainability relationship. On one side, the fire forecasts directly serve SDG 13 (Climate Action) by enabling proactive intervention before fires ignite, and SDG 15 (Life on Land) by protecting the Congo Basin and savanna ecosystems from uncontrolled burning. On the other hand, the compression and carbon analysis demonstrate that the AI system itself has been made environmentally responsible: the most compressed LSTM delivers the best accuracy per gram of carbon emitted of any configuration tested, proving that sustainability and performance reinforce rather than undermine each other.

7.2 Limitations

Several limitations should be acknowledged:

- **Tested on Central Africa only:** All models were built and evaluated exclusively on Central Africa. The system cannot simply be transferred to Australia, the Mediterranean, or Amazonia without retraining, because the vegetation types, fire seasons, and climate drivers differ fundamentally between regions.
- **No weather data included:** The models use only vegetation cover and calendar month as inputs. Temperature, rainfall, humidity, and wind speed all critical fire triggers are not included. Adding weather data would likely improve accuracy considerably, particularly for unusual fire seasons caused by extreme drought or heat events.
- **Satellite data has blind spots:** The burned area data comes from a NASA satellite that can be obscured by cloud cover during the fire season, may miss small fires, and provides only monthly totals rather than day-by-day tracking. Any errors or gaps in the satellite record feed directly into the AI's training and limit its accuracy ceiling.
- **Limited training data for the map-reading model:** The Swin Transformer was trained on only 47 monthly landscape snapshots. This type of model normally benefits from thousands of training images. Although we took precautions against overfitting and the results were strong, the model would likely improve further with a longer satellite record.
- **Six-year training window may miss long-term trends:** Training on 2017–2022 captures the most recent fire patterns but may miss longer-term shifts in fire behaviour driven by decades of deforestation, land-use change, or gradual climate warming. Fire seasons in the early 2000s may have had different dynamics that the model has not learned.
- **LSTM loses accuracy when heavily compressed:** When the LSTM is maximally compressed using precision reduction, its accuracy drops from 84% to 73% fire explanation, a meaningful loss for high-stakes decisions. Deployment on the most severely hardware-constrained devices would require accepting this trade-off, or adopting a more advanced compression approach (see future work below).

7.3 Future Work

Several extensions would strengthen and expand this work:

- **Add weather data:** Including temperature, rainfall, humidity, and wind alongside vegetation data would give the model the complete set of fire conditions fuel, weather, and ignition risk and is expected to substantially improve forecasts during unusual fire seasons.
- **Location-level predictions from the map model:** Adding a decoder layer to the Swin model would allow it to produce detailed hectare-level fire severity maps rather than a single regional figure, enabling direct comparison with the LSTM and making its outputs more actionable for ground teams.

- **Improved cross-patch communication in the map model:** Enabling the Swin to share information across the boundaries between landscape tiles rather than within each tile only would improve its ability to model fire spread across connected landscapes, particularly at the forest–savanna boundary.
- **Adapting the system for other fire-prone regions:** Taking the trained LSTM and retraining it on a small amount of data from Australia, the Mediterranean, or Amazonia would test whether the fire patterns it has learned in Central Africa translate to other regions and whether the physical common-sense rules remain valid across different biomes.
- **Smarter compression to preserve LSTM accuracy:** Training the LSTM with compression awareness built in from the start rather than compressing it after training would likely reduce the accuracy drop from 84% to 73% substantially, enabling a smaller, faster model without the current accuracy compromise.
- **Teaching a smaller model from a larger one:** Using the LSTM to train a much simpler, smaller model similar to an experienced teacher training a student [7] could produce a highly compact system that retains most of the LSTM's accuracy at a fraction of the size, offering a complementary route to field-ready deployment.

7.4 Conclusion

This project shows that AI can deliver practical, trustworthy wildfire forecasts that directly support both SDG 13 (Climate Action) and SDG 15 (Life on Land). The compressed time-memory model under 1 MB in size, physically constrained so it can never produce impossible predictions, is ready to be deployed on affordable hardware in the field. If rolled out across Central Africa's fire-prone savanna belt, such a system could give fire management agencies the advance warning they need to shift from fighting fires after they start to preventing the worst of them before they ignite. The cost in both lives, ecosystems, and carbon emissions of that shift could be enormous. Three independent lines of evidence point to the same practical conclusion: invest in monitoring and protecting broadleaved forest edges and shrubland zones. These are where the fires start, where the predictions are most reliable, and where early action has the greatest return. Responsible AI, accurate, explainable, physically grounded, and energy-efficient is not a future aspiration for climate action in Central Africa. It is available now.

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